

WEEK 3 - PRACTICE RESOURCES

Topic: Statistics & Predictive Modelling with Python

1. Statistical Foundations & Hypothesis Testing

- Khan Academy - Statistics and Probability - <https://www.khanacademy.org/math/statistics-probability>
- Scipy.stats Documentation & Examples - <https://docs.scipy.org/doc/scipy/reference/stats.html>
- Real Python - Python Statistics Fundamentals - <https://realpython.com/python-statistics/>
- GitHub - Statistical Analysis with Python (Thomas Haslwanter) - https://github.com/thomas-haslwanter/statsintro_python
- YouTube - StatQuest: Statistics Fundamentals - <https://www.youtube.com/playlist?list=PLblh5JKOoLUK0FLuzwntyYI10UQFUhsY9>

2. Scikit-learn Fundamentals & Machine Learning Workflow

- Scikit-learn Official Tutorial - <https://scikit-learn.org/stable/>
- Scikit-learn User Guide - https://scikit-learn.org/stable/user_guide.html
- GitHub - Scikit-learn Tutorial (Andreas Mueller) - https://github.com/amueller/introduction_to_ml_with_python
- Kaggle Learn - Introduction to Machine Learning - <https://www.kaggle.com/learn/intro-to-machine-learning>
- YouTube - Machine Learning Course by Andrew Ng - https://www.youtube.com/watch?v=vStJoetOxJg&list=PLkDaE6sCZn6FNC6YRfRQc_FbeQrF8BwGI

3. Practical Exercises in Python – Linear & Logistic Regression

- Scikit-learn Linear Regression Examples - https://scikit-learn.org/stable/modules/linear_model.html
- GitHub – Logistic Regression with Real Dataset <https://github.com/susanli2016/Machine-Learning-with-Python/blob/master/European%20Soccer%20Regression%20Analysis%20using>

[%20scikit-learn.ipynb](#)

- StatsModels Documentation (for statistical interpretation) - <https://www.statsmodels.org/stable/regression.html>

4. Model Evaluation & Performance Metrics

- Scikit-learn Model Evaluation Guide - https://scikit-learn.org/stable/modules/model_evaluation.html
- Confusion Matrix Visualisation Tools - https://scikit-learn.org/stable/auto_examples/model_selection/plot_confusion_matrix.html
- ROC Curve and AUC Explained - <https://machinelearningmastery.com/roc-curves-and-precision-recall-curves-for-classification-in-python/>
- Cross-Validation Techniques - https://scikit-learn.org/stable/modules/cross_validation.html
- YouTube – Confusion Matrix Explained Visually <https://www.youtube.com/watch?v=Kdsp6soqA7o>
- Medium – Choosing the Right Metric <https://medium.com/@kevinrioharris/beyond-accuracy-choosing-the-right-metric-to-evaluate-machine-learning-models-4dd6a0cd12c8>

5. Hands-On with scikit-learn

- Official scikit-learn Tutorials – Regression & Classification https://scikit-learn.org/stable/supervised_learning.html

6. Public Datasets for Practice

- UK Crime Statistics - <https://data.police.uk/data/>
- Companies House Data - https://download.companieshouse.gov.uk/en_output.html
- London Transport Data (TfL) - <https://tfl.gov.uk/info-for/open-data-users/>
- Netflix Viewing Activity (*useful for churn prediction exercises*) <https://www.kaggle.com/datasets/shivamb/netflix-shows>

7. Advanced ML Topics & Hyperparameter Tuning

- Scikit-learn GridSearchCV Documentation - https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.GridSearchCV.html
- Hyperparameter Tuning with Random Search - https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.RandomizedSearchCV.html
- Feature Selection Techniques - https://scikit-learn.org/stable/modules/feature_selection.html
- Pipeline Creation for ML Workflows - <https://scikit-learn.org/stable/modules/compose.html>
- GitHub - Hyperparameter Tuning Examples - <https://github.com/automl/auto-sklearn>

8. Platforms for Interactive Practice

- Google Colab – <https://colab.research.google.com>
- Kaggle Notebooks – <https://www.kaggle.com/code>
- Deepnote (EU-hosted) – <https://deepnote.com>
- Binder – <https://mybinder.org>

9. Learning Communities & Forums

- Stack Overflow – Python tag <https://stackoverflow.com/questions/tagged/python>
- Kaggle Discussion Forums <https://www.kaggle.com/discussion>
- r/datascience <https://www.reddit.com/r/datascience/>
- Data Science Stack Exchange <https://dataengineering.wiki/Concepts/Concepts>